



Praktikum Design of Wind Farms

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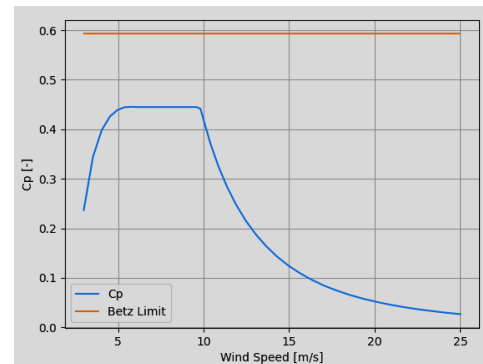
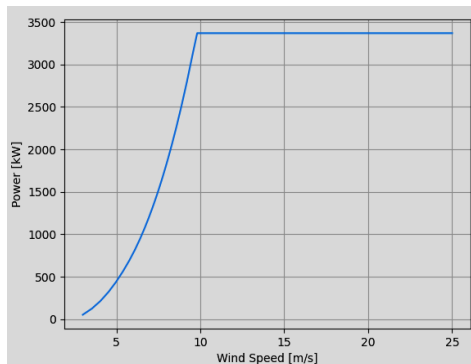
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Assignment 1 (5%): Basics and Recapitulation (Wind Conditions, Weibull Distribution, and AEP)

This assignment aims at introducing students to the analysis of wind conditions and the estimation of wind turbine annual energy production (AEP).

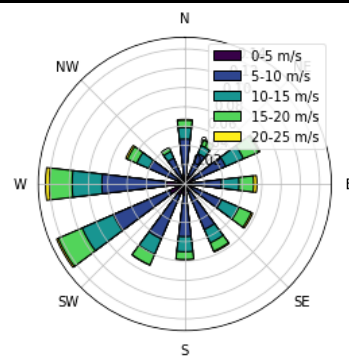
For a given wind turbine installed at a specific location, key parameters are as follows:

Item	Value	Units
Name	IEA 3.4MW Reference	N/A
Rated Power	3370	kW
Rated Wind Speed	9.8	m/s
Cut-in Wind Speed	3	m/s
Cut-out Wind Speed	25	m/s
Rotor Diameter	130	m
Hub Height	120	m
Control	Pitch Regulated	N/A



A .csv file is available on [GitHub](#).

Item	Value	Units
Target wind direction	240	[°]
C (scale parameter of the Weibull function)	7.5	[m/s]
k (shape parameter of the Weibull function)	2.72	[-]
Total turbine working hours in this condition	1270	hrs



wind rose

1. Calculate the AEP and capacity factor (CF) of the wind turbine for the given wind direction. Please also upload your calculation file (Excel, Matlab, etc.).